

Smooth finite-dimensional Banach spaces satisfy Property (P)

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Source question

This packet concerns arXiv:2407.07490, Debmalya Sain, Arpita Mal, Kalidas Mandal and Kallol Paul, *On uniform Bishop-Phelps-Bollobas type approximations of linear operators and preservation of geometric properties*. In Section 3 the authors define Property (P) and ask:

Is it true that each finite-dimensional Banach space X satisfies Property (P)? What about the infinite-dimensional case?

either $\|x_0 - h_0\| < \epsilon$ or $\|x_0 + h_0\| < \epsilon$, which contradicts the choice of ϵ . Therefore, the pair $(\mathbb{X}, \mathbb{X})$ does not have the uniform sBPBp with respect to $\mathcal{F}$. $\square$

We end this section with an open question that we could not answer. In view of the results obtained here, it is apparent that an answer to the following question will be useful in having a better understanding of the uniform ϵ -BPB approximations of operators as well as the norm attainment set of an operator.

Question: Is it true that each finite-dimensional Banach space $\mathbb{X}$ satisfies Property (P)? What about the infinite-dimensional case?

4. APPLICATIONS OF UNIFORM ϵ -BPB APPROXIMATION ON SOME PARTICULAR BANACH SPACES

In this section we deal with operators on some well studied Banach spaces like ℓ_∞^n and ℓ_1^n , in view of the preservation of the norm attainment set under uniform ϵ -BPB approximations. We note that in case of norm one operators which are not extreme contractions, the situation is completely understood by virtue of Theorem 3.4. Therefore, we restrict ourselves mostly to the case of extreme contractions. We begin with extreme contractions in $\mathbb{K}(\ell_\infty^n, \ell_\infty^n)$. First we require the following

The source paper proves Property (P) for finite-dimensional polyhedral spaces, for Hilbert spaces, and later for ℓ_p^2 when $p > 2$ is an integer. The result below gives a new partial answer for all finite-dimensional smooth real Banach spaces.

Definitions

Let X be a real Banach space. For $A \in \mathcal{L}(X)$, set

$$M_A = \{x \in S_X : \|Ax\| = \|A\|\}.$$

The space X has *Property (P)* if there is an $r_0 > 0$ such that for every norm-one non-isometry $A \in \mathcal{L}(X) \setminus \text{Iso}(X)$, there is $x_A \in S_X$ with

$$B(x_A, r_0) \cap M_A = \emptyset.$$

Equivalently, the norm attainment set of a norm-one non-isometry can never be uniformly dense in the unit sphere.

Recall also that a finite-dimensional real Banach space is smooth when every $x \in S_X$ has a unique norming functional $j(x) \in S_{X^*}$. In finite dimension this implies that the norm is Frechet differentiable on $X \setminus \{0\}$, with derivative at $x \in S_X$ equal to $j(x)$, and the map $x \mapsto j(x)$ is continuous on S_X .

Proof intuition

Suppose Property (P) fails. Then there are norm-one non-isometries A_n whose norm-attainment sets M_{A_n} are $1/n$ -dense in S_X . Such an operator almost preserves the norm of every unit vector, so compactness forces A_n close to the compact isometry group.

Conjugate A_n by a nearest isometry and write the result as

$$B_n = I + t_n C_n, \quad t_n = \|B_n - I\| > 0.$$

After passing to a subsequence, $C_n \rightarrow C$ with $\|C\| = 1$. Contractivity of B_n gives $j(x)(Cx) \leq 0$. The density of M_{B_n} gives the opposite inequality, hence

$$j(x)(Cx) = 0 \quad (x \in S_X).$$

That identity says C is an infinitesimal isometry: $\exp(sC)$ is an isometry for every $s \in \mathbb{R}$. But then the isometry $\exp(t_n C)$ is closer to $B_n = I + t_n C_n$ than I is, contradicting the choice of the nearest isometry.

Main theorem

Theorem 1. *Every finite-dimensional smooth real Banach space satisfies Property (P).*

Proof. Let X be finite-dimensional and smooth. Denote by $G = \text{Iso}(X)$ the group of surjective linear isometries of X . Since X is finite-dimensional, G is a compact subset of $\mathcal{L}(X)$.

Assume, toward a contradiction, that X fails Property (P). Then for each n there exists a norm-one non-isometry $A_n \in \mathcal{L}(X)$ such that M_{A_n} is $1/n$ -dense in S_X .

We first note that $\text{dist}(A_n, G) \rightarrow 0$. Indeed, for every fixed $x \in S_X$ choose $y_n \in M_{A_n}$ with $\|x - y_n\| \leq 1/n$. Since $\|A_n\| = 1$,

$$1 \geq \|A_n x\| \geq \|A_n y_n\| - \|A_n\| \|x - y_n\| \geq 1 - \frac{1}{n}.$$

Thus any operator-norm cluster point A of the sequence A_n satisfies $\|Ax\| = 1$ for all $x \in S_X$, hence belongs to G . Compactness of the unit ball of $\mathcal{L}(X)$ proves $\text{dist}(A_n, G) \rightarrow 0$.

For each n , choose $U_n \in G$ with $\|A_n - U_n\| = \text{dist}(A_n, G)$, and put

$$B_n = U_n^{-1} A_n.$$

Then B_n is a norm-one non-isometry, $M_{B_n} = M_{A_n}$, and

$$\text{dist}(B_n, G) = \|B_n - I\|.$$

The last equality follows because left multiplication by U_n^{-1} is an isometry of $\mathcal{L}(X)$ and maps G onto G .

Set

$$t_n = \|B_n - I\| > 0, \quad C_n = \frac{B_n - I}{t_n}.$$

After passing to a subsequence, $C_n \rightarrow C$ in $\mathcal{L}(X)$, with $\|C\| = 1$. We now show that

$$j(x)(Cx) = 0 \quad (x \in S_X).$$

First fix $x \in S_X$. Since $\|B_n\| \leq 1$,

$$\|x + t_n C_n x\| = \|B_n x\| \leq 1 = \|x\|.$$

Dividing by t_n and using Frechet differentiability of the norm at x , with $C_n x \rightarrow Cx$, gives

$$j(x)(Cx) \leq 0. \tag{1}$$

For the reverse inequality, choose $y_n \in M_{B_n}$ with $\|y_n - x\| \leq 1/n$. Then $y_n \rightarrow x$, and

$$\|y_n + t_n C_n y_n\| = \|B_n y_n\| = 1 = \|y_n\|.$$

The norm is differentiable uniformly on compact subsets of $X \setminus \{0\}$ in finite dimension, and $C_n y_n \rightarrow Cx$. Therefore

$$0 = \lim_n \frac{\|y_n + t_n C_n y_n\| - \|y_n\|}{t_n} = j(x)(Cx). \tag{2}$$

Equations (1) and (2) give the desired identity.

We claim that $\exp(sC) \in G$ for every $s \in \mathbb{R}$. Fix $x \neq 0$ and set $g(s) = \|\exp(sC)x\|$. The vector $\exp(sC)x$ is never zero. If $v_s = \exp(sC)x / \|\exp(sC)x\|$, then differentiability of the norm and the identity $j(v)(Cv) = 0$ on S_X give

$$g'(s) = j(v_s)(C \exp(sC)x) = \|\exp(sC)x\| j(v_s)(Cv_s) = 0.$$

Hence $g(s) = g(0) = \|x\|$ for all s . Thus $\exp(sC)$ is a linear isometry of X , and in finite dimension it is surjective, so $\exp(sC) \in G$.

Finally, since $\exp(t_n C) \in G$,

$$t_n = \text{dist}(B_n, G) \leq \|B_n - \exp(t_n C)\|.$$

But $B_n = I + t_n C_n$, $C_n \rightarrow C$, and

$$\exp(t_n C) = I + t_n C + O(t_n^2),$$

so

$$\|B_n - \exp(t_n C)\| \leq t_n \|C_n - C\| + O(t_n^2) = o(t_n).$$

This contradicts $t_n > 0$. Therefore X must satisfy Property (P). □

Scope and limitations

This is a partial answer to the source question. It proves the finite-dimensional smooth real case. It does not settle arbitrary finite-dimensional norms, because nonsmooth nonpolyhedral spaces do not have a global single-valued differentiable duality map. It also does not settle the infinite-dimensional part of the source question; the proof uses compactness of both the operator unit ball and the isometry group.

The argument should be adaptable to complex spaces if smoothness is formulated as real Frechet differentiability of the norm and $j(x)$ is interpreted as the real derivative. This packet records only the real version to avoid hiding that convention.

Verification notes

- The failure of Property (P) is used exactly as the existence of norm-one non-isometries whose M_A 's are arbitrarily dense in S_X .
- The passage from dense M_{A_n} to closeness to the isometry group uses only finite-dimensional compactness.
- The nearest-isometry conjugation is essential: it gives $\text{dist}(B_n, G) = \|B_n - I\|$, so the later exponential isometry contradicts nearest-point minimality.
- The only differentiability input is standard for finite-dimensional smooth real Banach spaces: the norm is Frechet differentiable off the origin, with continuous derivative on the sphere.
- A bounded novelty check was performed against the run indexes for arXiv:2407.07490, Property (P), uniform BPB/Bishop-Phelps-Bollobas keywords, and nearby BPB phrases. No duplicate packet or exact later answer was found.

Human review recommendation

Review as a likely valid partial result. The main line to audit is the standard differentiability fact for finite-dimensional smooth real Banach spaces and its use with $y_n \rightarrow x$ and $C_n y_n \rightarrow Cx$. Once that lemma is accepted, the compact Lie-tangent contradiction is quite rigid.

Reference

D. Sain, A. Mal, K. Mandal and K. Paul, *On uniform Bishop-Phelps-Bollobas type approximations of linear operators and preservation of geometric properties*, arXiv:2407.07490.